TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Verification of Ohm's Law Using a Tangent Galvanometer

No ammeter: the current is read from the deflection of a compass needle caught between two magnetic fields — the earth's and the coil's — and V/I is shown to stay constant as the current changes.

PHY-1204 · PHYSICS-II SESSIONAL · EXPERIMENT 5 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to verify Ohm's law — that the potential difference across a conductor is proportional to the current through it — measuring the current with a tangent galvanometer. The instrument is a vertical circular coil of n turns and radius r, set in the magnetic meridian, with a small compass needle pivoted at its centre. Current in the coil makes a field at the centre, $B = \mu_0 nI/2r$, directed perpendicular to the earth's horizontal field B_h. The needle settles along the resultant, deflected by θ from the meridian, where the **tangent law** holds:

$$\tan\theta = B / B_h \quad \rightarrow \quad I = (2r\,B_h / \mu_0\,n)\tan\theta = K\tan\theta$$

The constant K is the **reduction factor** of the galvanometer — the current that deflects the needle by 45°. With the galvanometer in series with a specimen resistor, a rheostat and a cell, and a voltmeter across the specimen, each setting of the rheostat gives a pair (V, $I = K\tan\theta$). If V/I is the same for every setting, $V \propto I$: Ohm's law is verified, and the constant ratio is the specimen's resistance.

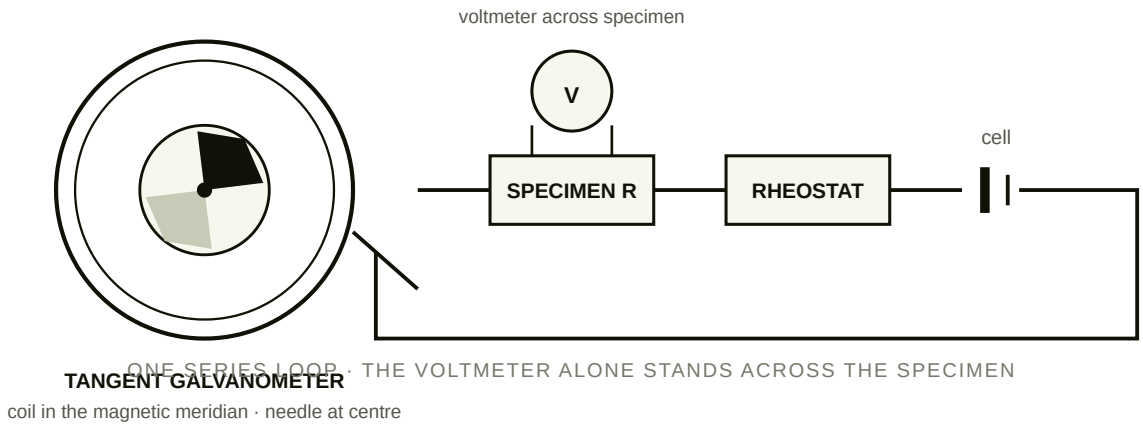


Figure 1 — The circuit: cell, rheostat, specimen and tangent galvanometer in series; the voltmeter across the specimen only. The rheostat changes the current; the needle reports it through tan θ.

2. Apparatus and procedure

- **Apparatus:** tangent galvanometer, specimen resistor (a few ohms), voltmeter, rheostat, cell, commutator (or reversing key), connecting wires.
- Set the galvanometer: level it, rotate the coil into the magnetic meridian so the needle reads 0–0 with no current, and use a chosen number of turns (n = 50 here). Compute $K = 2rB_h/(\mu_0 n)$ for the local B_h.
- Connect the series loop (Fig. 1) with the voltmeter across the specimen. Keep all other magnets, iron and current-carrying leads well away from the galvanometer.
- Set the rheostat for the largest safe current. Read θ (both ends of the needle; average) and V. Reverse the current with the commutator and read θ again — averaging direct and reversed deflections cancels zero and alignment errors.
- Reduce the current in steps (six or more settings), recording θ and V each time. Tabulate $I = K \tan\theta$ and V/I.
- Plot V against I: Ohm's law predicts a straight line through the origin, of slope R.

3. Worked calculation

Local $B_h = 3.6 \times 10^{-5}$ T; coil r = 0.10 m, n = 50 turns:

$$K = (2 \times 0.10 \times 3.6 \times 10^{-5})/(4\pi \times 10^{-7} \times 50) = 0.115 \text{ A}$$

Θ (mean, direct & reversed)	tan Θ	I = K tan Θ (A)	V (volt)	V/I (Ω)
30°	0.577	0.066	0.266	4.0

45°	1.000	0.115	0.460	4.0
60°	1.732	0.199	0.797	4.0

V/I is constant at the specimen's resistance, 4.0 Ω, while the current trebles — Ohm's law holds, and the V–I graph is a straight line through the origin.

4. Precautions

- The tangent galvanometer is a magnetic instrument in a magnetic world: no bar magnets, iron masses, phones or spare current leads anywhere near it; the connecting wires to it should be twisted together so their fields cancel.
- Deflections should stay between about 30° and 60°: below that, reading error in θ looms large; above it, tan θ races away and the same error explodes in I.
- Read both ends of the needle and average; reverse the current and average again.
- Do not let the specimen warm up — its resistance is the thing under test, and a warm specimen bakes a curve into the "straight" line.

Practice (viva preparation)

1. Derive $I = K \tan\theta$ from the fields at the coil's centre, defining K.
2. A tangent galvanometer ($r = 9$ cm, $n = 10$, $B_h = 3.6 \times 10^{-5}$ T) deflects 45°. What current flows?
3. Why is the coil set in the magnetic meridian before starting?
4. The V–I plot is straight but does not pass through the origin. Suggest an experimental cause.
5. Why are deflections near 80° avoided even though tan θ is then large and "sensitive"?

Answer key

1. The coil's field at its centre, $B = \mu_0 nI/2r$, is perpendicular to B_h when the coil stands in the meridian. The needle aligns with the resultant, so $\tan\theta = B/B_h = \mu_0 nI/(2rB_h)$; rearranging, $I = (2rB_h/\mu_0 n)\tan\theta$, and the bracket is K, the current for $\theta = 45°$.
2. $I = K \tan 45° = K = 2 \times 0.09 \times 3.6 \times 10^{-5}/(4\pi \times 10^{-7} \times 10) = 6.48 \times 10^{-6}/1.257 \times 10^{-5} \approx 0.52$ A.
3. Only then is the coil's field exactly perpendicular to B_h, which the tangent law assumes; a coil at an angle mixes a component along B_h and the simple tan θ relation fails.
4. A zero error: the needle not reading zero at no current (coil slightly out of the meridian / bent pointer), so every computed I carries a constant offset — or a voltmeter zero error doing the same to V. Reversing the current and averaging both needle ends is the cure.

5. Because $I \propto \tan\theta$, a fixed reading error $\delta\theta$ produces $\delta I/I = \delta\theta/(\sin\theta\cos\theta)$, which is smallest at 45° and grows sharply toward 0° and 90°. At 80° a half-degree misreading costs several percent of I.
